

Concept Learning versus Problem Solving: A Cognitive Difference

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An area of informative and insightful research in chemical education over the past 25 years has been comparing student performance on algorithmic versus conceptual problems. An algorithmic problem is one that can be solved using a memorized set of procedures; a conceptual problem requires the student to work from an understanding of a concept to the solution to the problem, where no memorized procedure is likely to be known to the student. The findings across these studies showed that the majority of students in high school and college chemistry courses rely almost exclusively on an algorithmic approach to quantitative problem solving (1–5). The results of this approach to “learning” chemistry are counterproductive for students’ longer-term needs from their chemistry coursework because, among other issues, experience with an algorithmic approach does not facilitate conceptual understanding (6).

Furthermore, learning algorithms as the primary mode of problem solving is contrary to the goals of presenting chemistry as a process of scientific inquiry and promoting students’ intellectual development. The *National Science Education Standards* defines “scientific inquiry” (7, p 214) as:

[A] set of interrelated processes by which scientists and students pose questions about the natural world and investigate phenomena; in doing so, students acquire knowledge and develop a rich understanding of concepts, principles, models, and theories.

Memorization of an algorithm is neither posing questions about the natural world nor investigating phenomena. The *Standards* also say (7, p 173) that:

[F]or students to develop the abilities that characterize science as inquiry, they must actively participate in scientific investigations, and they must actually use the cognitive and manipulative skills associated with the formation of scientific explanations.

The development of cognitive skills is an essential component of a student’s education, yet memorization of algorithms has no effect on these skills.

The development of cognitive skills—particularly those used in scientific reasoning—should be a central goal for any high school or college chemistry course (8). One reason for this is that students with poor reasoning skills cannot solve conceptual problems (9). This lack of skill leaves a student with

no choice except to memorize algorithms if they want to survive a chemistry course. This is a significant problem for a large percentage of students who enroll in chemistry courses. Studies of the reasoning skills of students entering college consistently show that only about 25% are well equipped with the skills needed to solve conceptual problems (10–12). Our own research-in-progress data show that only 8% of high school juniors and 22% of college freshmen who are beginning a chemistry course enter with sufficient reasoning skills to have the potential to be successful conceptual problem solvers.

Thus, we hypothesized that a *cause* of the gap between conceptual and algorithmic problem-solving ability is poor reasoning skills. To test this hypothesis, we partially replicated and also extended the line of research on algorithmic versus conceptual problem solving of Nurrenbern (13), Nurrenbern and Pickering (14), Sawrey (15), Sanger (16), Sanger, Campbell, Felker, and Spencer (17), and Sanger and Phelps (18). The key results from these studies indicated that:

1. Students perform significantly better on algorithmic questions than conceptual questions that test understanding of gas properties and stoichiometry at the particulate level
2. The algorithmic–conceptual difference in performance exists with all students, even those who rank in the top fraction of the class
3. This type of investigation has a critical dependence on the questions used as measuring instruments

Theoretical Framework

The theory of learning that provides a framework for this study is Piaget’s constructivism (19, 20, 21). As Piaget described (22, p 15):

[H]uman knowledge is essentially active. To know is to assimilate reality into systems of transformations.... Knowledge, then, is a system of transformations that become progressively adequate.

In other words, individuals structure their knowledge uniquely, yet there is a common *process* by which human knowledge develops. Part of this common process is the development of the ability to reason systematically and by imagining hypothetical outcomes beyond those that are currently observable (23).

However, development of good reasoning skills requires deliberate practice, so although all high school and college students without learning disabilities have the physiological potential to reason at the level required to learn abstract scientific concepts, not all students practice sufficiently to develop the necessary skills (24–26).

Methods

Algorithmic and conceptual problem-solving ability was measured with questions from Nurrenbern and Pickering (14) and other questions with similar characteristics. Four algorithmic–conceptual question pairs were administered, one pair on each of the four midterm examinations in a first-semester general chemistry course. The topics and sources of these questions are summarized in Table 1; complete questions are in the online supplement. In general, the algorithmic questions were similar to questions discussed in class and assigned as homework. The conceptual questions were based on principles covered in the course, although they were truly “problems” in that students had no opportunity to solve a similar question in class or in homework before seeing the question on a midterm examination.

Table 1. Distribution of Questions Used in the Study

Exams	Concept Tested	Question Sources	Question Type	Question Summaries
1	Density	(27)	Algorithmic	Given density, convert mass to volume
		(28)	Conceptual	Liquid displacement, varying density ^a
2	Stoichiometry	(14)	Algorithmic	Limiting reactant ^b
		(14)	Conceptual	Particulate-level reaction and equation
3	Ideal Gas	(14)	Algorithmic	$PV = nRT$, solve for one variable
		(14)	Conceptual	Particulate-level change with change in temperature of a gas
4	Molarity	Authors	Algorithmic	Given molarity, density, and volume, calculate mass
		Authors	Conceptual	Particulate-level ratio from molar concentration

^aWe used two questions to assess students' conceptual understanding of density; one where same-sized balls of different density are dropped into the same liquid, and one where same-sized balls of the same density are dropped into liquids of different density. A complete understanding of the density concept yields success on both questions.

^bThe algorithmic stoichiometry question in Nurrenbern and Pickering (14) is multiple choice. On our instrument, students were required to generate a solution to the question rather than pick an answer from a set of choices.

Scientific reasoning skill was measured with the Classroom Test of Scientific Reasoning, multiple-choice version (revised edition, August 2000) (29–30). The skills assessed by this instrument are conservation of weight, conservation of displaced volume, proportional thinking, identification and control of variables, probabilistic thinking, correlational thinking, and hypothetico-deductive reasoning (29).

Instruction in the first-semester general chemistry course was based on a combination of active learning techniques, including guided inquiry and a modified peer-led team learning (PLTL) (31) approach. Required course materials included an unpublished active-learning, guided-inquiry textbook and a parallel in-class workbook, both developed by one of the authors of this paper (MSC). A different author was the instructor for the course (JCD). Each week, students attended three (50-minute) lectures, one (three-hour) laboratory, and one (two-hour) PLTL workshop. The lectures began with a 10-minute quiz based on the homework assigned from the textbook to complement the material from the previous lecture, and the remaining 40 minutes were a combination of lecture and active learning in which the peer leaders guided groups of eight–ten students working on problems from the in-class workbook. The laboratory component of the course was inquiry-based (32, 33), and the PLTL workshop component consisted of the same group of students and the same peer leader as in lecture, using materials from that project (34). The problem-solving ability questions were included as questions on the four midterm exams in the course, administered in the 4th, 7th, 11th, and 14th weeks of the 15-week semester.

Scoring

Algorithmic Questions

Each algorithmic question could be solved through application of a set of memorized procedures, without the need for conceptual understanding, so there was no reason to assess partial understanding. These questions were scored as correct (1) or incorrect (0). Otherwise correct solutions that contained minor errors such as incorrect significant figures were scored as correct for this study.

Conceptual Questions

The conceptual density question was coded with a scheme based on that in Westbrook and Marek (35): four points for complete understanding, three for partial understanding, two for partial understanding with specific misconception, one for specific misconception, and zero for no understanding. The score from each of the two parts was then considered; 3 or 4 on each part was counted as correct (1), and other scores were incorrect (0). The other conceptual questions were scored as correct (1) or incorrect (0) based on student explanations.

Classroom Test of Scientific Reasoning

This instrument consists of 11 double-multiple choice questions and 2 single multiple-choice questions. The double multiple-choice questions require students to answer a question and give an explanatory reason for their answer. Both parts must be correct for credit. This yields a maximum score of 13.

Subjects

The study's subjects were 94 students who were enrolled in a first-semester general chemistry course at a doctoral-granting public university with a 96% undergraduate acceptance rate and an enrollment of about 13,000 students. This institution enrolls approximately two-thirds of the total annual enrollment in general chemistry in a Fall–Spring offering and one-third in a Spring–Summer offering of the two-semester sequence. The subjects were enrolled in the Spring–Summer sequence in 2002. Based on initial course enrollment data, 79% of these students had one or more years of high school chemistry. The most common declared majors were prepharmacy, biology, and geology; 60% of the students identified one of these three fields as their major.

Results

Summary Statistics

Table 2 summarizes student success on each of the four question pairs. Since the stoichiometry and ideal gas question pairs were repeated from the Nurrenbern and Pickering (14) and Sawrey (15) studies, a comparison of student success rates across the three studies is illustrated in Figures 1 and 2. The results are generally consistent across the three studies: success on algorithmic questions is always significantly higher than on conceptual questions covering the same subject matter.

The average score on the Classroom Test of Scientific Reasoning (29) was 7.2 (out of 13 possible), and the standard deviation was 2.9. In general, a score of about 9 or higher indicates sufficient scientific reasoning skill to have the underlying potential to be a successful conceptual problem solver (9, 36). It is readily apparent from these data that the average general chemistry student lacks the requisite skills needed for success in solving conceptual problems.

Comparative Statistics

To test our hypothesis that poor reasoning skills are a cause of the gap between conceptual and algorithmic problem-solving ability, we divided the subjects into groups based on their scientific reasoning test scores to compare the performance of the more skilled reasoners, which we defined as the top one-third, with the poorer reasoners, defined as the bottom one-third in the class. This comparison method is analogous to Sawrey's (15) upper and lower 27% of the class, and it clearly separates the sample into distinct groups based on reasoning ability. The mean score on the Classroom Test of Scientific Reasoning of the top one-third of the subjects initially enrolled in the course was 10.5, and the mean score for the bottom one-third of students was 3.7.

Table 3 summarizes comparisons of the better reasoners with the poorer reasoners on each question plus their performance on the ACS First-Term General Chemistry Exam,¹ which we used as the final exam in the course. The better reasoners outperformed the poorer reasoners on each question, both algorithmic and conceptual, as well as on the ACS exam. A statistically significant difference was found on one of four algorithmic questions and three of the four conceptual questions and on the ACS exam. The mean ranking of the better reasoners on the 70-item ACS exam was the 91st percentile, and the poorer reasoners

averaged at the 57th percentile (for all subjects, the mean and median percentile scores were 74 and 80, respectively).

In comparing the results between the better and poorer reasoners on the four pairs of questions, the same pattern emerges. The percent success for each question declines in the order:

better reasoners, algorithmic → poorer reasoners, algorithmic → better reasoners, conceptual → poorer reasoners, conceptual

Table 2. Comparative Rates of Student Success on the Four Algorithmic–Conceptual Question Pairs

Question Category	N	Algorithmic Avg. Scores, %	Conceptual Avg. Scores, %
Density	90	61.1	41.1
Stoichiometry	82	62.2	9.8
Ideal gas	77	83.1	42.9
Molarity	71	60.6	22.5

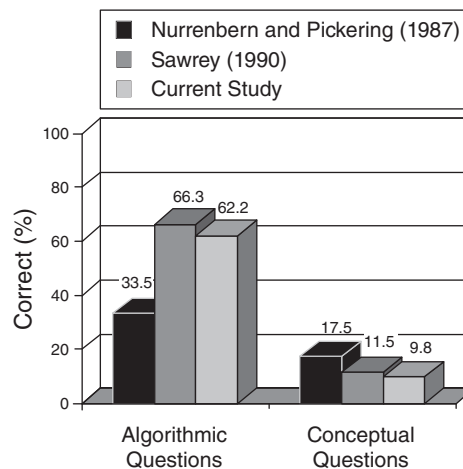


Figure 1. Comparison of students' results on stoichiometry questions across three studies.

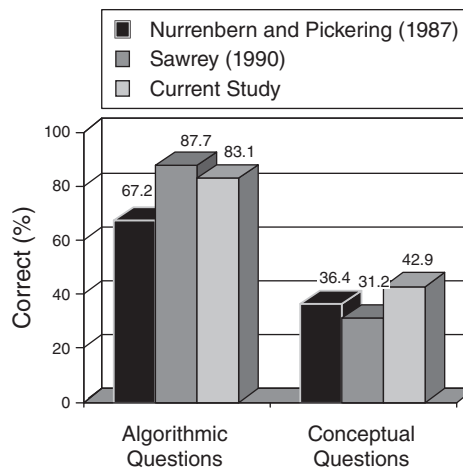


Figure 2. Comparison of results of gas laws questions across three studies.

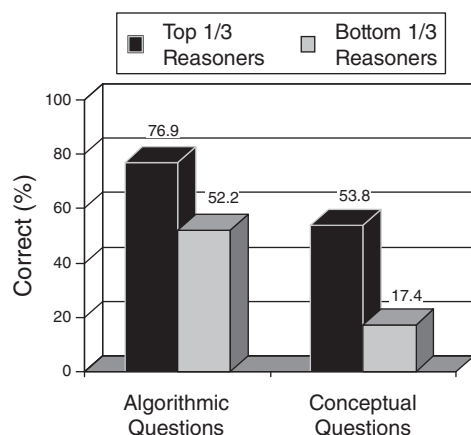


Figure 3. Comparison of the results on the density questions for each reasoning skill group.

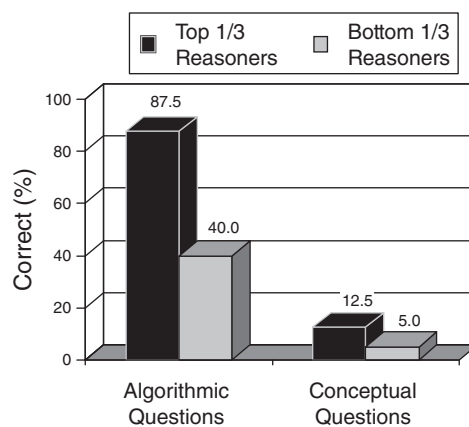


Figure 4. Comparison of the results on the stoichiometry questions for each reasoning skill group.

Figures 3–6 illustrate these differences in reasoning skills on the four pairs of questions. The results on the conceptual molarity question were most striking: 47% of the better reasoners succeeded in solving the problem, but none of the poorer reasoners was successful. These figures show that although algorithmic and conceptual problem-solving ability is sensitive to the question used as a measuring instrument, the general pattern is consistent

across all questions. Given that the problem-solving success rate for better reasoners is significantly higher on three of the four conceptual questions but only one of the four algorithmic questions, the data indicate that variation in scientific reasoning skills is a cause of the gap between algorithmic and conceptual problem-solving ability. Reasoning ability difference was also a cause of the difference in performance on the ACS exam.

Table 3. Comparing Performance of Students Classified as Better and Poorer Reasoners

Concepts Tested	Question Types	Reasoning Levels	N	Mean Scores	Standard Deviation Values	p-Values
Density	Algorithmic	Top One-Third	26	0.77	0.43	0.082
		Bottom One-Third	23	0.52	0.51	
	Conceptual	Top One-Third	26	0.54	0.51	0.016
		Bottom One-Third	23	0.17	0.39	
Stoichiometry	Algorithmic	Top One-Third	24	0.88	0.34	0.001
		Bottom One-Third	20	0.40	0.50	
	Conceptual	Top One-Third	24	0.13	0.34	0.614
		Bottom One-Third	20	0.05	0.22	
Gas laws	Algorithmic	Top One-Third	20	0.90	0.31	0.065
		Bottom One-Third	19	0.63	0.50	
	Conceptual	Top One-Third	20	0.55	0.51	0.048
		Bottom One-Third	19	0.21	0.42	
Molarity	Algorithmic	Top One-Third	19	0.63	0.50	0.506
		Bottom One-Third	16	0.50	0.52	
	Conceptual	Top One-Third	19	0.47	0.51	0.001
		Bottom One-Third	16	0.00	0.00	
ACS Final	Raw Scores	Top One-Third	18	56.50	6.537	0.000
		Bottom One-Third	16	42.563	11.355	
ACS Final	Percentiles	Top One-Third	18	91		
		Bottom One-Third	16	57		

Notes: Bold type indicates $p < 0.05$; the Fisher exact probability test was used to generate p -values for the questions; a t -test was used to analyze the ACS final.

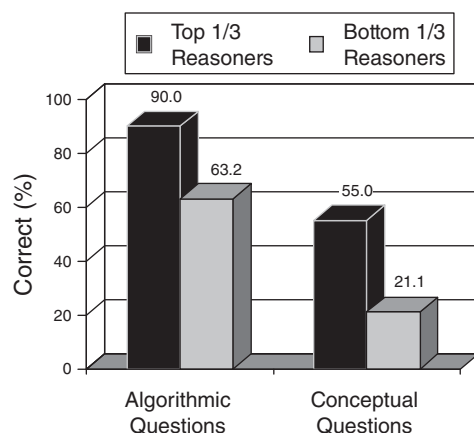


Figure 5. Comparison of the results on the gas laws questions for each reasoning skill group.

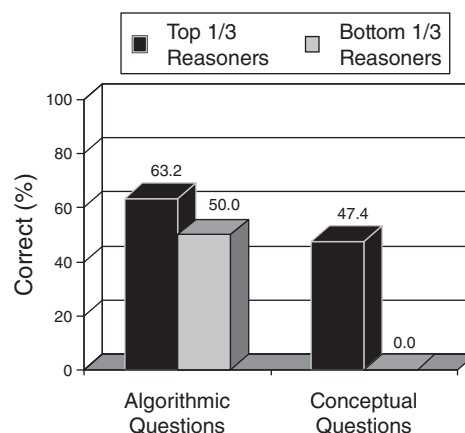


Figure 6. Comparison of the results on the molarity questions for each reasoning skill group.

Conclusions

In 1993, Nakhleh (37) asked, "Are our students conceptual thinkers or algorithmic problem solvers?" The data from this study indicate that a significant fraction of our students have no choice other than to be algorithmic problem solvers because their reasoning skills are not sufficiently developed to allow them to successfully solve conceptual problems.

It is also notable that textbook authors and chemistry instructors have responded to the results of relatively early studies such as Nurrenbern and Pickering (14) by placing an increased emphasis on helping students make macroscopic-particulate connections (38). The course in which the subjects of this study were enrolled utilized a textbook rich in particulate-level illustrations, and a variety of modern, research-based instructional strategies were integrated into the curriculum. In spite of the improved curriculum, the algorithmic-conceptual gap across the group of studies investigating this topic remains approximately the same over a number of decades. Bodner (39) suggested that changing the curriculum may not be enough, proposing that the methods of curriculum delivery must also be changed. We agree, and we also propose that an increased emphasis on the development of reasoning skills must be integrated into science curricula from middle school through college. As predicted from Piaget's constructivism, there are cognitive differences in the underlying reasoning abilities among students in science courses. These differences prevent a significant fraction of students from being successful conceptual problem solvers.

Implications for Instructors

It is not a simple task to change science curricula so that the focus shifts more toward fostering intellectual development, yet it is possible. An excellent long-term model that can be followed was developed by Adey and Shayer (40). They showed that by introducing science lessons specifically designed to develop students' reasoning skills over two years in middle school, permanent improvements in these skills relative to a control group were measurable three years after the intervention (41-44). Not only were students' abilities in science and mathematics improved, but also in English, providing evidence that improve-

ment in scientific reasoning skills leads to increased academic performance that transfers across disciplines.

Although the duration of an intervention is a critical criterion, instructors can facilitate the development of students' reasoning skills in a semester- or year-long course by increasing the emphasis on requiring students to solve problems slightly beyond their current abilities (45). This approach is effective, in part, because most students will only develop their skills if they are required to do so. Curricula based on instructional strategies such as the learning cycle (32, 46) and guided inquiry (47, 48) are excellent course designs that embed repeated opportunities for students to develop their reasoning skills.

Note

1. We used the ACS Examinations Institute General Chemistry (First-Term) 2002 exam version. <http://www4.uwm.edu/chemexams/materials/exams.cfm> (accessed Mar 2008).

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Supplement

The set of algorithmic–conceptual question pairs used in this study on general chemistry topics of density, stoichiometry, gas laws, and molarity